

# **Cross-Subject EEG Transfer Learning Using Riemannian Geometry, Tangent Space Alignment, and Spatial–Riemannian Feature Fusion**

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## **Author Note**

This report was prepared as an independent computational neuroscience and brain–computer interface research project investigating subject-independent EEG decoding using geometric machine learning methods.

## Abstract

Brain-computer interfaces based on electroencephalography remain constrained by extensive subject-specific calibration requirements. This study evaluates four transfer-learning strategies under leave-one-subject-out cross validation: baseline Riemannian decoding, subject-aligned Riemannian decoding, an EEGNet-inspired deep-learning baseline, and an RCSP–Riemannian feature fusion framework. The fusion model achieved the strongest generalization performance ( $61.5\% \pm 8.1\%$ ), outperforming both geometric and deep-learning baselines and demonstrating the importance of covariance geometry for cross-subject decoding.

## Introduction

Practical deployment of motor-imagery BCIs requires models capable of generalizing to unseen users without lengthy calibration sessions. Inter-subject variability in anatomy, cortical organization, skull conductivity, and oscillatory dynamics creates substantial distributional shifts between participants. Traditional Euclidean feature extraction methods frequently fail under these conditions. Riemannian geometry provides an alternative representation by treating covariance matrices as points on the symmetric positive definite manifold, preserving their intrinsic geometric structure during learning and classification.

## Materials and Methods

Experiments were conducted using the EEG Motor Movement/Imagery Dataset. Nine participants performing left- and right-hand motor imagery tasks were included. EEG recordings were standardized, mapped to standard 10–05 electrode positions, bandpass filtered between 8 and 30 Hz, and epoched from one to four seconds following cue presentation. Covariance matrices were estimated using Oracle Approximating Shrinkage. Performance was evaluated using strict leave-one-subject-out validation in which each participant served as a completely unseen test subject.

## Model Architectures

Four model families were evaluated. The baseline model projected covariance matrices into tangent space and classified them using logistic regression. The aligned model introduced subject-specific covariance recentering prior to tangent-space projection. A lightweight EEGNet-inspired convolutional architecture was implemented as a deep-learning baseline. Finally, a feature-fusion model combined regularized common spatial pattern features with tangent-space representations and classified them using a linear support vector machine.

## Results

Baseline Riemannian decoding achieved 54.8% accuracy. Subject-specific alignment improved performance to 59.8%, suggesting that covariance shifts explain a substantial proportion of

inter-subject variability. EEGNet achieved 49.1%, performing near chance level under leave-one-subject-out evaluation. The RCSP–Riemannian fusion framework achieved the highest performance at  $61.5\% \pm 8.1\%$ , indicating that discriminative spatial filtering and covariance geometry capture complementary information relevant to motor imagery decoding.

## Discussion

Several findings emerge from these experiments. First, subject-independent EEG decoding remains substantially more difficult than within-subject classification. Second, geometric methods consistently outperformed the deep-learning baseline, reinforcing previous literature demonstrating the effectiveness of covariance geometry for low-sample EEG applications. Third, subject-specific alignment improved transfer performance, supporting the hypothesis that covariance shifts are a primary source of inter-subject variability. Finally, the superior performance of the fusion framework suggests that spatial and geometric features encode complementary information and should be jointly exploited in future BCI systems.

## Limitations

The present study focused exclusively on binary motor imagery classification and did not investigate multi-class paradigms, cross-session variability, or real-time deployment. The dataset size was relatively small for deep-learning approaches and likely contributed to the poor EEGNet performance. Furthermore, advanced transfer-learning techniques such as Tangent Space Alignment, domain adversarial training, and transformer architectures were not implemented.

## Future Directions

Future work will investigate cross-session transfer learning, cross-montage adaptation, domain adaptation methods, online decoding with Lab Streaming Layer, and transformer-based architectures for EEG representation learning. The eventual objective is deployment within subject-independent neuroprosthetic and neurotechnology systems.

Table 1. Model performance summary.

| model                    | mean_accuracy      | std_accuracy       | 95% Confidence Interval |
|--------------------------|--------------------|--------------------|-------------------------|
| Baseline Riemannian      | 0.5481481481481482 | 0.087488976377909  | $\pm 0.171$             |
| Aligned Riemannian       | 0.5975308641975309 | 0.1151726800404904 | $\pm 0.226$             |
| Simple EEGNet            | 0.491358024691358  | 0.0878019969713311 | $\pm 0.172$             |
| RCSP + Riemannian Fusion | 0.6148148148148148 | 0.0808901098808946 | $\pm 0.159$             |

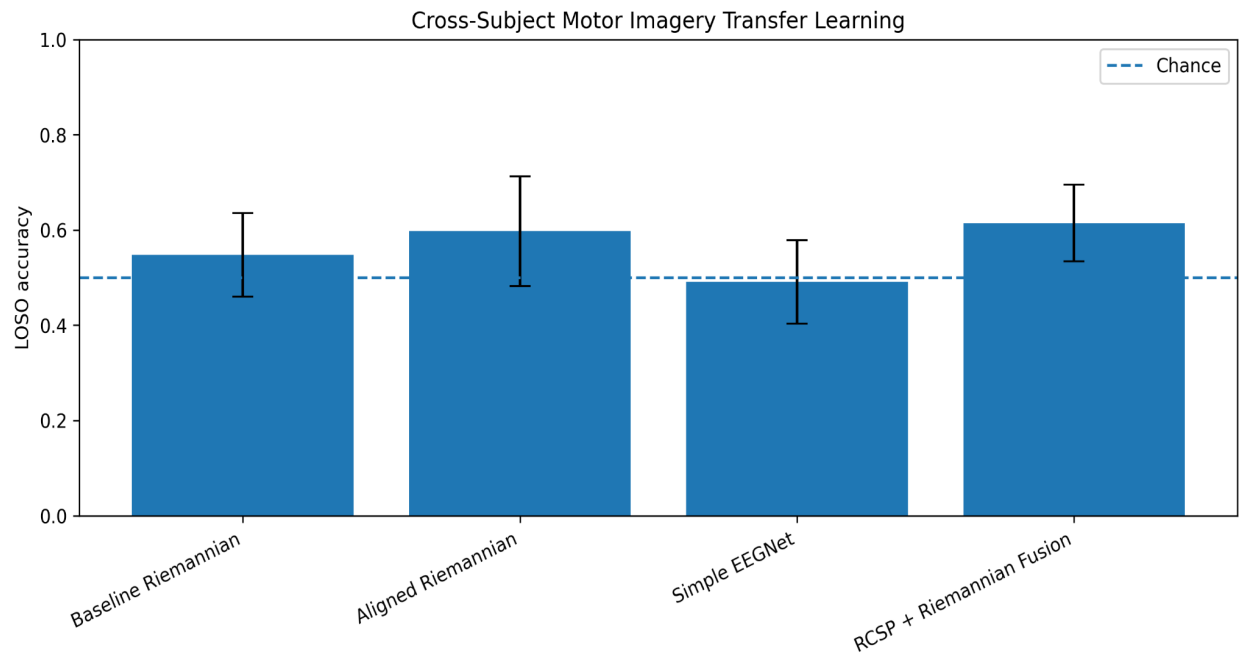


Figure 1. Overall comparison of transfer-learning models under LOSO evaluation.

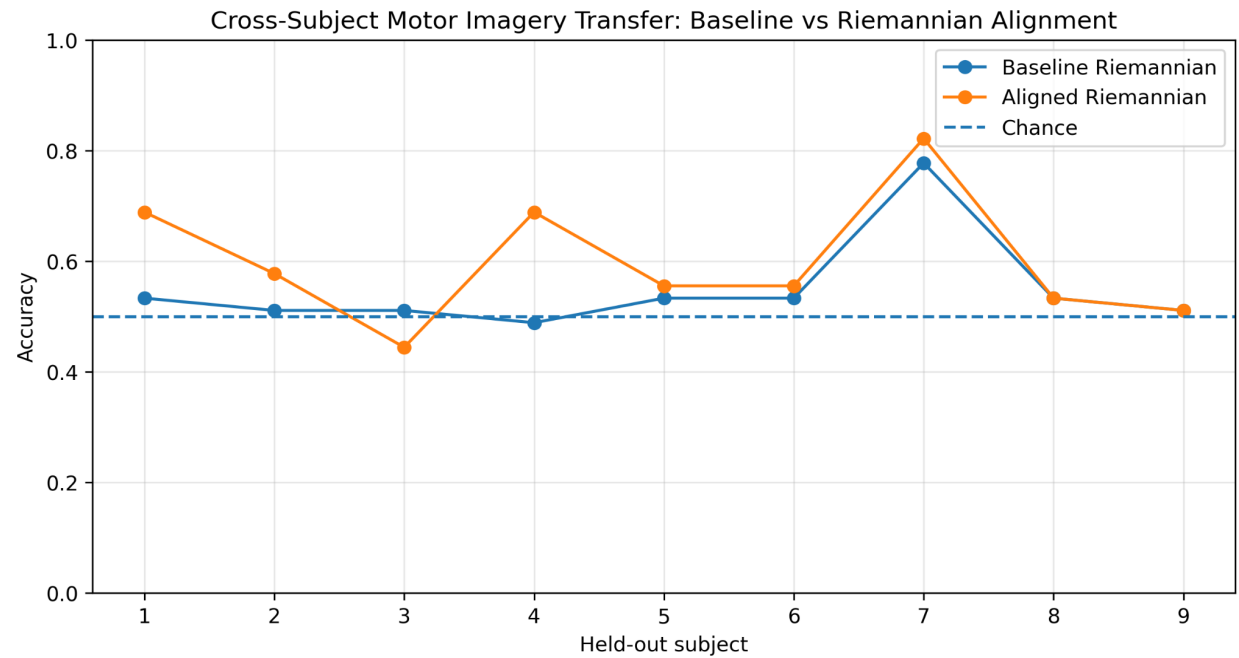


Figure 2. Subject-wise comparison of baseline and aligned Riemannian decoders.

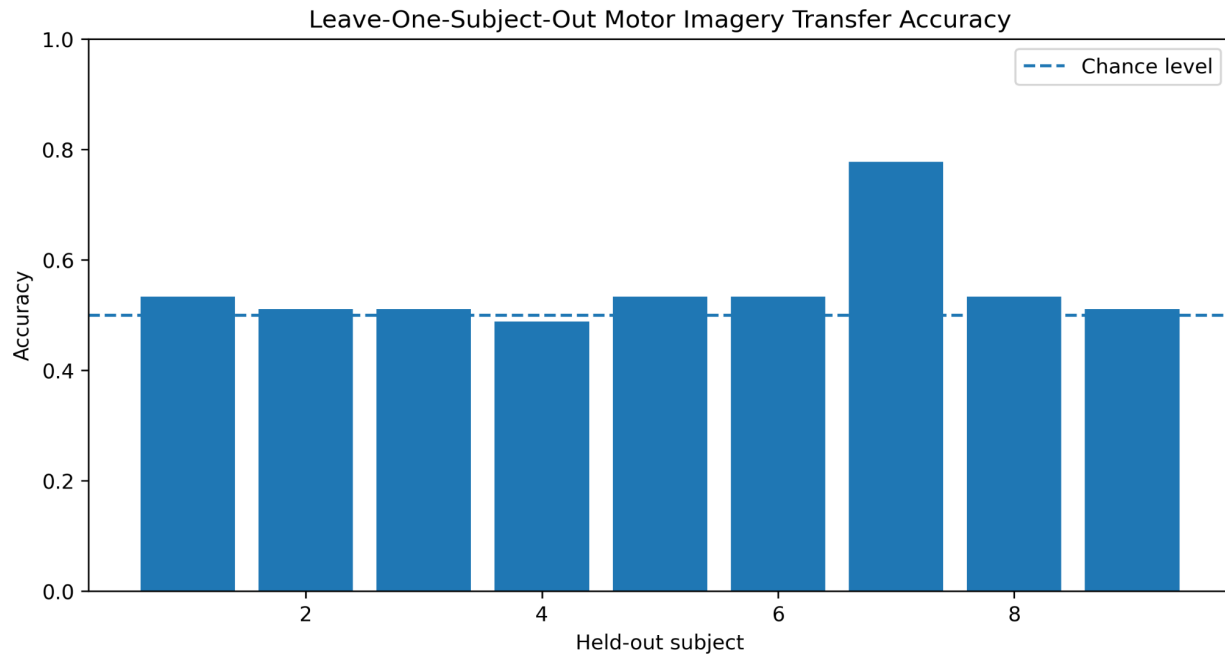


Figure 3. Baseline Riemannian LOSO performance across held-out subjects.



Figure 4. Schematic overview of the transfer-learning pipeline.

## References

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